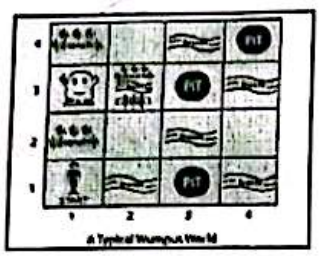
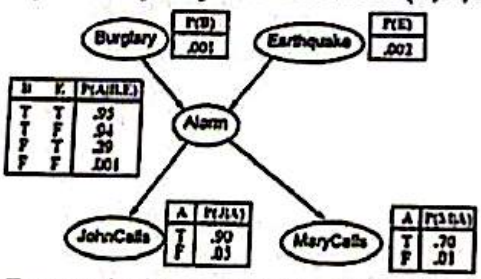
International Islamic University Chittagong
Department of Computer Science & Engineering
B. Sc. in CSE Semester Final Examination, Autumn-2025
Course Code: CSE-3635 Course Title: Artificial Intelligence
Total marks: *50* Time: *2* hours *30* minutes

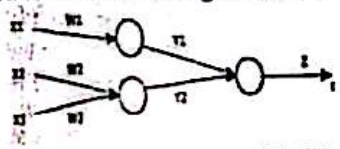
[Answer *all* the questions. Figures in the right hand margin indicate full marks. Separate answer script must be used for Group A and Group B]

Group-A

			CLO	Marks
1.	a)	 Consider the following sentences in the Wumpus world above: R1: ¬P1,1 R2: B1,1 ↔ (P1,2 v P2,1) R3: B2,1 ↔ (P1,1 v P2,2 v P3,1) R4: ¬B1,1 R5: B2,1 KB containing R1 through R5, and show how to prove P3,1 that is, there is pit in [3,1].	CLO2	6
1.	b)	Consider the following axioms: P ..... (1) (P ^ Q)→R .....(2) (S v T)→Q .....(3) T ——————(4) a) Convert the formulas into CNF. b) Prove that whether R is true or not using any inference technique.	CLO2	4
		OR		
1.	a)	Convert the following sentences to propositional logic: i) If it is morning, then the birds are singing. ii) I am eating if and only if I am hungry.	CLO2	1+1
1.	b)	Consider the following statements- • John likes all kind of food. • Apple and vegetable are food. • Anything anyone eats and not killed is food. • Anil eats peanuts and still alive. • Harry eats everything that Anil eats. Using the Resolution technique, prove the statement "John Likes Peanuts".	CLO2	6
1.	c)	Discuss two differences between forward chaining and backward chaining.	CLO2	2
2.	a)	In a medical diagnosis system, the probability that a patient has Disease D is 0.002. The probability that a patient shows symptom S given the disease is 0.95, and the probability of showing the symptom without the disease is 0.04. Calculate the probability that the patient actually has the disease given that the symptom is observed.	CLO3	4
2.	b)	A Bayesian network showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J, and M stand for Burglary, Earthquake, Alarm, John Calls, and Mary Calls respectively. The independent conditional probability helps us to write in a simplified way the joint distribution P(H, E, A, I, M)  i) Express the joint distribution P(H, E, A, J, M) in terms of the conditional probabilities (and independences) expressed in the Bayesian Network above. ii) Probability of the event that the alarm has sounded but neither a burglary nor an earthquake has occurred and both Mary and John call.	CLO3	6

iii) Probability of the event that the alarm has sounded and Burglary has occurred, an earthquake has not occurred, and both Mary and John call.

Group-B

3:	a)	Explain Bayes Theorem. Clearly define prior probability, likelihood, and posterior probability.	CLO3	3
3	b)	Given the following membership function for the fuzzy set "Warm Temperature": $\mu_{Warm}(x) = \begin{cases} 0, & x \le 15 \\ \frac{x-15}{10}, & 15 < x \le 25 \\ 1, & x > 25 \end{cases}$ (a) Explain what a membership function is in fuzzy logic. (b) Find the degree of membership of a temperature of 20°C in the fuzzy set "Warm".	CLO3	4
3.	c)	In a city, 60% of drivers use mobile phones while driving. The probability of an accident when a driver uses a phone is 0.05, and when not using a phone is 0.01. If an accident has occurred, calculate the probability that the driver was using a mobile phone.	CLO3	3
4.	a)	Contrast expert systems and neural networks in terms of knowledge representation, knowledge acquisition and explanation. Give one domain in which the expert system approach would be more promising and one domain in which neural network approach would be more promising.	CLO3	4
4.	b)	A software company is developing an AI-powered customer support chatbot for an online shopping platform. The chatbot must understand customer messages, track conversation context, and respond politely even when customers are indirect or emotional. Answer the following: i. Explain Natural Language Processing (NLP) and justify why NLP is necessary for this chatbot. ii. Write down both the top down and bottom up parse tree for the sentence: "The chatbot resolved the issue".	CLO3	6
	OR	Using the following simple grammar for a fragment of English. S → NP VP; NP → Det N; VP → V NP; Det → the; N → chatbot; N → issue; V → resolved; V → handled; NP → N; VP → V		
5.	a)	A neuron in an ANN uses a sigmoid activation function: $f(x) = 1 / (1 + \exp(-x))$ a) Name one similarity and one difference between artificial and biological neural networks. b) Explain how the sigmoid function helps in deciding between two classes in binary classification. c) If the input to the neuron is x = 1, find the sigmoid output.	CLO4	2+1+1
5.	b)	Review the following neural network and compute the output Z.  The input values are X1=15, X2=8, X3=14 and the corresponding weights are: W1=0.6, W2=0.3, W3=0.1, Y1=0.6, Y2=0.45. a) Compute the value of Z without using any transfer (activation) function. b) Apply a threshold at all neurons. If the value is 5 or less, output 0; otherwise, output 1. Re-calculate Z. c) Compute the value of Z assuming that the ReLU transfer function is used at all neurons.	CLO4	2+2+2
	OR			
5.	a)	What is semi-supervised learning? Explain the concept gradient descent optimization.	CLO4	1+3
5.	b)	Consider a single neuron with two inputs X1 and X2, and output Y, learning the logical AND operation. The truth table is given below.	CLO4	6

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

Develop an ANN model using Delta rule, Where,
Threshold, T = 0.5; Learning rate, alpha = 0.2; Initial weights W1=0.1 and W2=-0.5

International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE, Final Examination, Spring 2025

Course Code: CSE-3635 **Course Title:** Artificial Intelligence

Total Marks: 50 **Time:** 2 Hours 30 Minutes

[Answer *all* the following questions. Figures in the right hand margin indicate full marks. Use a separate answer script for Group-A and Group-B.]

Group A

			Marks	COs
1.	a)	Define Inference. Determine whether the following sentence is i) Satisfiable ii) Contradictory iii) Valid $(P \vee Q) \rightarrow (P \wedge Q)$	3	2
	b)	State modus ponens with an example? Consider following sentences: **Premise1:** Every Bangladeshi likes Jackfruit. **Premise2:** Shihab is a Bangladeshi. **Conclusion:** Shihab likes Jackfruit. Why Propositional logic fails here?	3	2
	c)	Given knowledge base R1, R2, R3, R4, R5 below, show that P1,2 is false. R1: $\neg P1,1$ R2: $B1,1 \leftrightarrow (P1,2 \vee P2,1)$ R3: $B2,1 \leftrightarrow (P1,1 \vee P2,2 \vee P3,1)$ R4: $\neg B1,1$ R5: $B2,1$	3	2
	d)	What is the logical translation of the following statement? "None of my friends are perfect." (A) $\exists x(F(x) \wedge \neg P(x))$ (B) $\exists x(\neg F(x) \wedge P(x))$ (C) $\exists x(\neg F(x) \wedge \neg P(x))$ (D) $\neg\exists x(F(x) \wedge P(x))$	1	2
		OR		
1.	a)	A statute mandates that it is an offense for a citizen of Nation X to furnish controlled substances to a nation designated as a prohibited recipient. The nation of Zargon is formally designated as a prohibited recipient by the government of Nation X. Zargon has recently operationalized a novel, high-capacity power generation system. All essential rare-earth elements required for the construction of this power generation system were supplied by Ms. Anya, who holds citizenship in Nation X. Prove by that Ms. Anya Sharma has committed an offense by both - 1) Forward chaining 2) Backward chaining	6	2
	b)	Describe the following concepts with example- 1) Generalized Modus Ponens 2) Hypothetical Syllogism	4	2
2	a)	Why does first-order predicate logic fail to adapt to a domain like medical diagnosis? How can agents manage the uncertainty raised by a lack of environmental knowledge?	3	3

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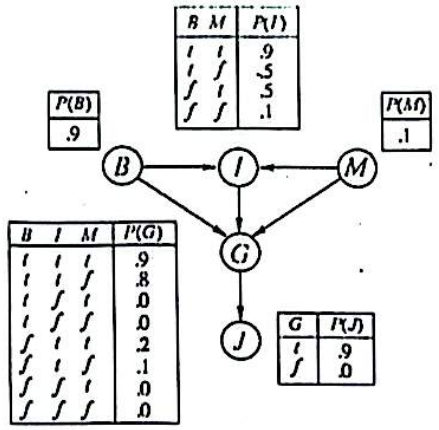
b) A domain consisting of just the three Boolean Variables Toothache, Cavity and Catch. The full joint distribution is a 2 x 2 x 2 table as shown in figure. 2 3

	Toothache		¬Toothache	
	Catch	¬Catch	Catch	¬Catch
Cavity	0.108	0.012	0.072	0.008
¬Cavity	0.016	0.064	0.144	0.576

Calculate:

i. P(Cavity v ¬Catch)
ii. P(¬Cavity|Toothache, ¬Catch)
iii. P(Cavity|¬Toothache)
iv. P(Toothache|Cavity)

c) A Bayesian network with Boolean variables B = BrokeElectionLaw, I = Indicted, M = PoliticallyMotivatedProsecutor, G= FoundGuilty, J = Jailed, showing both the topology and the conditional probability tables. 5 3



I) Calculate the value of P(b, i, m, g, j)
II) Calculate the probability that someone goes to jail given that they broke the law, have been indicted, and face politically motivated prosecutor.

Group B

3. a) Define conditional probability. Write the difference between degree of belief and degree of truth with example. 3 3

b) "Fuzzy set is the generalized version of crisp set"- Do you agree with the statement? Justify your answer 2 3

c) Given the following membership functions for the fuzzy set "Temperature": 2 3

$$\mu_{Cold}(x) = \begin{cases} 1, & x \leq 10 \\ \frac{20-x}{10}, & 10 < x < 20 \\ 0, & x > 20 \end{cases}$$

Find the degree of membership for a temperature of 15°C in the set "Cold".

d) A doctor knows that meningitis causes the patient to have a stiff neck 50% of the time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is 1/50,000, and the prior probability that any patient has a stiff neck is 1/20. Find the probability of patients with stiff neck having Meningitis. 3 3

OR

3. a) Describe the concept of inductive learning with an example. Is reinforcement learning considered a subset of inductive learning? Justify your answer. 4 3

b) Write down both the top down and bottom up parse tree for the sentence: "Nusair created the program" using the following simple grammar for a fragment of English in Figure 1. 6 3

S → NP VP

NP → the NP1

NP → PRO

NP → PN

NP → NP1

NP1 → ADJS N

ADJS → ε | ADJS ADJ

VP → V

VP → V NP

N → file | program

PN → Nusair

PRO → ε

ADJ → short | long | fast

V → printed | created | wanted

Figure 1: A simple grammar for a fragment of English

4. a) Write down the steps through which NLP is conducted? Why is sequencing important in NLP? 3 4

b) What are the key components of an expert system? Explain with a diagram. 3 3

c) Imagine you want to create an 'Intelligent Recipe Recommender' system. This system should act like an expert chef, suggesting recipes and giving cooking advice. How would you use Natural Language Processing (NLP) and Expert System ideas together to build this? Explain each step. 4 3

5 a) What is supervised learning? Explain the concepts of forward propagation and back propagation in neural networks. 4 4

b) Consider a single neuron with two inputs X1 and X2, and output Y, learning the logical AND operation. The truth table is given below. 6 4

x_1	x_2	y
0	0	0
0	1	0
1	0	0
1	1	1

Develop an ANN model using Delta rule, Where,
Threshold, T = 0.2; Learning rate, alpha = 0.2; Initial weights W1=0.3 and W2=-0.1

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International Islamic University Chittagong

Department of Computer Science & Engineering

B.Sc. in CSE, Final Examination, Autumn 2024

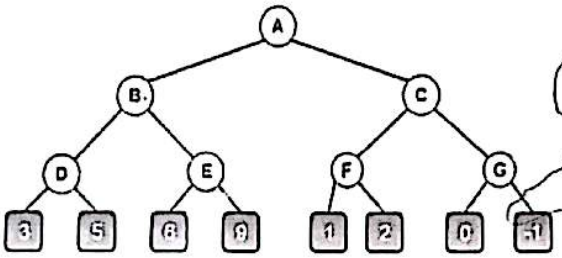
Course Code: CSE-3635 **Course Title:** Artificial Intelligence

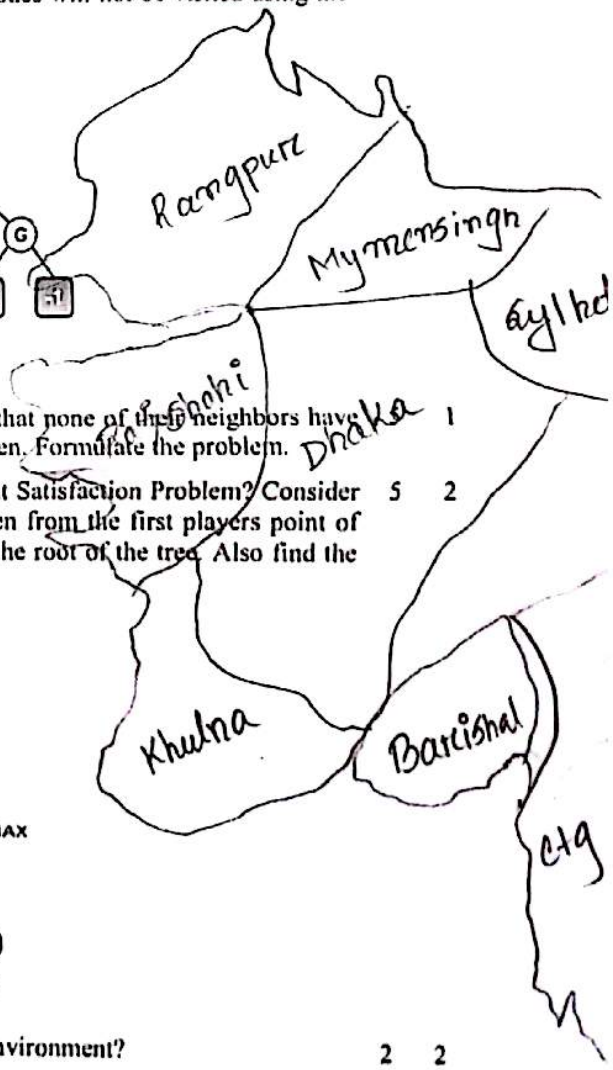
Total Marks: 50 **Time:** 2 Hours 30 Minutes

[Answer *all* the following questions. Figures in the right hand margin indicate full marks. Use a separate answer script for Group-A and Group-B.]

Group A

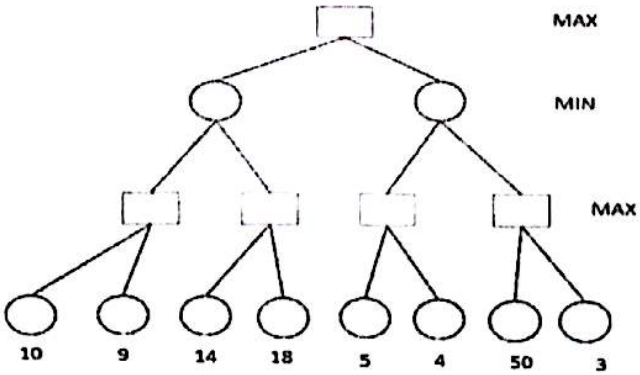
				COs
1.	a)	What is CSP? Do you consider map coloring as an example of CSP? If so, identify the variables, domains and constraints associated with this problem	5	1
	b)	What is an imperfect decision in terms of game playing? Consider the following game tree where static scores are all from the first player's point of view. In the tree, which nodes will not be visited using the alpha beta pruning procedure?	5	2





OR

1.	a)	Consider the Bangladeshi map below. You want to color divisions so that none of their neighbors have the same color. The colors you may choose from are Red, Blue and Green. Formulate the problem.	1	
	b)	What are the differences between Search-Based Problem and Constraint Satisfaction Problem? Consider the following two player game tree in which the static scores are given from the first players point of view. Apply the Minimax search algorithm and compute the value of the root of the tree. Also find the most convenient path for Max Node.	5	2



2	a)	How can the Wumpus World be described in terms of its features and environment?	2	2
	b)	*"As per the law, it is academic dishonesty for an IIUC student to cheat in the exam hall. During the AI exam, Rahim, a student of IIUC, has some Cheating materials, and all these materials were provided to him by Rafat, who is also an IIUC student."* Now, using forward and backward chaining prove that "Rafat is a dishonest student."	8	2

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Group B

3. a) A survey was conducted at a juice bar, and the following data were collected: 6 3

- 8 male students, 5 female students, 3 male teachers and 2 female teachers bought orange juice
- 6 male students, 7 female students, 2 male teachers and 4 female teachers bought mango juice
- 4 male students, 5 female students, 2 male teachers and 2 female teachers bought lemon juice

Answer the following question: C = teacher | student

i. Create a Join probability distribution table for the given data.
ii. P(C)=?
iii. P(C | Female)= ?
iv. P(C | Female, Student)= ?

Calculate the probability using inference by enumeration.

b) A well-known symptom of measles is a rash (the event of having which we denote R). Assume that the probability of having a rash if one has measles is P(R|M) = 0.95. However, occasionally children with flu also develop rash, and the probability of having a rash if one has flu is P(R|F) = 0.08. 4 3
Upon examining the child, the doctor finds a rash. What is the probability that the child has measles?

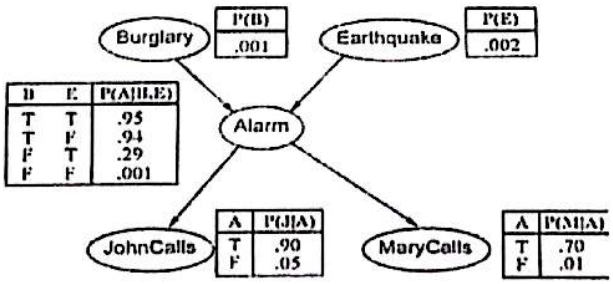
4. a) What is supervised learning? Briefly explain the concept of mathematical neuron using appropriate diagram. 4 4

b) Consider a single neuron with two inputs X1 and X2, and output Y, learning the logical OR operation. 6 4
The truth table is given below.

Case	X1	X2	Z
1	0	0	0
2	0	1	1
3	1	0	1
4	1	1	1

Develop an ANN model using Delta rule, Where,
Threshold, T = 0.5
Learning rate, alpha = 0.2
Initial weights W1=0.1 and W2=0.5

5. a) 3 3



Consider the Bayesian Network above:
Calculate the probability that alarm has sounded, but there is neither a burglary, nor an earthquake occurred, and John and Mary both called.

b) What is the probability that Mary call? 4 3

c) What is the role of membership function in Fuzzy Logic? Describe the fuzzy logic process diagram in brief. 3 3

OR

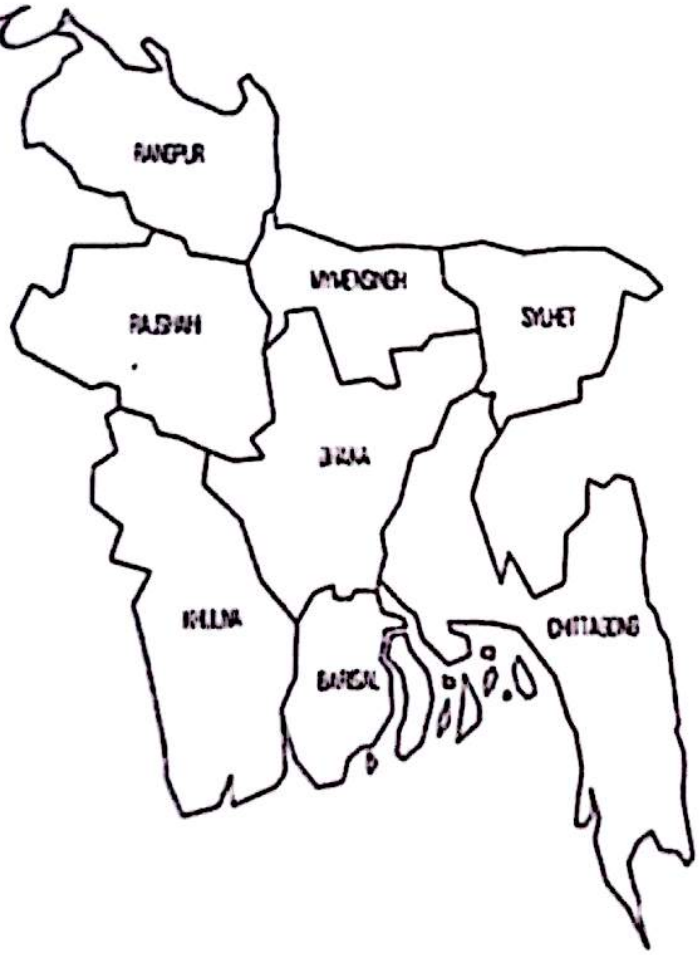
5 a) Compare top down and bottom up parsing approach in parsing sentence tree. 3 3

b) Convert A ⇔ (B ∨ C) into CNF. Prove that the contra positive of p → q is equal to p → q using truth table. 4 3

c) What do you mean by uncertainty? What are the different sources of uncertainty. 3 3

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1(a) OTC



International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Spring 2024

Course Code: CSE 3635 Course Title: Artificial Intelligence

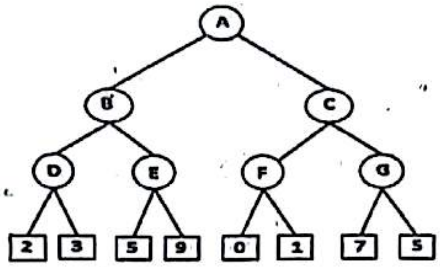
Total marks: 50

Time: 2 hours 30 minutes

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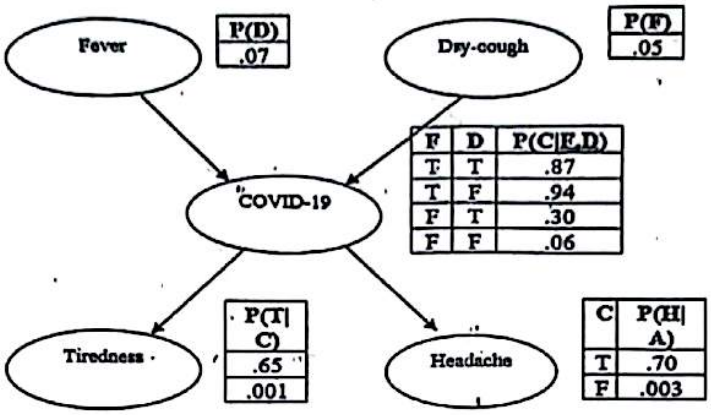
Course Outcomes and Bloom's Levels are mentioned in additional Column

Part A

			C	DL	Marks
1	a)	Consider the following game tree in which static scores are all from the first player's point of view:  i) Suppose the first player is the maximizing player. What move should be chosen? ii) In the game tree, what nodes would not need to be examined using the alpha-beta pruning procedure?	3	U	5
1	b)	What is CSP? Do you consider the map coloring as an example of CSP? If so, identify the variables, domains, and constraints associated with this problem.	3	An	5
		OR			
1	b)	Analysis between forward chaining and backward chaining. In what aspects forward chaining should be used?	3	An	5
2	a)	In a knowledge-based system, the following rules are given: Rule 1: A breeze in cell B22 implies that there is a pit in either cell P21, P23, P12, or P32. Rule 2: There is no breeze in cell B22. Using logical rules such as biconditional elimination, modus ponens, disjunction elimination, and other necessary logical rules, determine ¬ (P21 ∨ P23 ∨ P12 ∨ P32)	3	Ap	5
		OR			
2	a)	Consider the following sentences. i) All people who are graduating are happy ii) All happy people smile iii) Someone is graduating. iv) Is someone smiling? A) Convert the above sentences into FOL. B) Convert the sentences (FOL) into clausal form C) Prove iv) by using resolution.	3	Ap	2+2+

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2. b) The following Bayesian network, showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters C, F, D, T and H stand for COVID-19, Fever, Dry-Cough, Tiredness, and Headache respectively. The Independent conditional probability helps us to write in a simplified way the joint distribution P (C, F, D, T, and H). 3 Ap

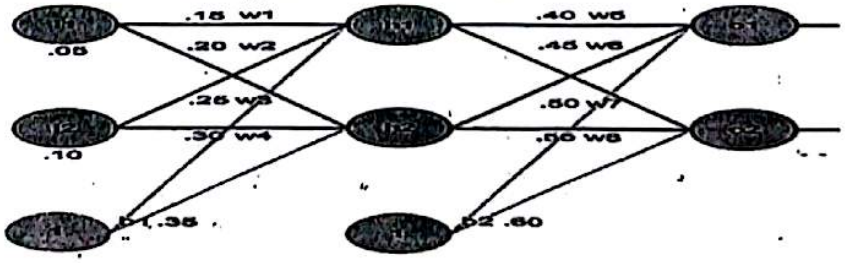


F	D	P(C|F,D)
T	T	.87
T	F	.94
F	T	.30
F	F	.06

(i) Express the joint distribution P (C, F, D, T, H) in terms of the conditional probabilities (and independencies) expressed in the Bayesian Network above.
(ii) Probability of the event that the patient is tested positive for COVID-19 but the patient does not have symptoms of neither a Fever nor Dry-cough and both symptoms of tiredness and headaches have occurred.

Part B

3	a)	Explain the Bayes Theorem.	4	U	3
3	b)	About 2/3 of drivers use their cell phone while driving. Suppose that you are 5 times more likely to get into an accident if you text and drive, and if you don't use your cell phone, you have a 1% chance of getting into an accident. What is the probability that someone was texting given that they got into an accident?	4	Ap	5
		OR			
3	b)	In a medical scenario, it is known that patients with dengue fever(+d) develop a rash(+r) 80% of the time. The doctor also has the following information: the prior probability of a patient having dengue fever is 1/10000 and the probability that a patient has rash given that they do not have dengue fever is 1/100. Find the probability that a patient has a dengue given that they have rash.	4	Ap	5
3	c)	What is NLP? Write down the steps through which NLP is conducted.	4	R	2
4	a)	How do neural networks differ from conventional computers? Briefly explain.	4	U	2
4	b)	Consider the following neural network model.	4	Ap	5

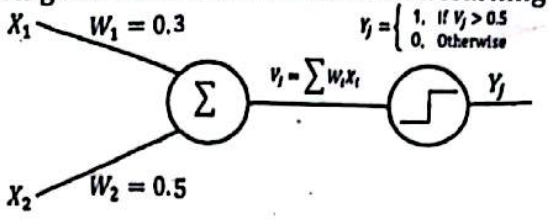


i) Compute the value of O1 and O2 without transfer function.
ii) Compute the value of O1 and O2 with sigmoid transfer function.

All error goes to next neuron.

OR

Consider the following neural network model with a learning rate of 0.2. 4 Ap 5



Demonstrate the learning process of the OR operation as shown below using the delta rule.

c) Considering the target output o1=0.01 and 02=0.99 in the network depicted in question 4(b), calculate the error for each output neuron as well as the total error of the network using the squared error function. 4 U 3

a) Write down the top down and bottom up parse tree for the sentence "Book the flight through Houston" using the following CFG. 4 Ap 4

Grammar	Lexicon
S → NP VP S → Aux NP VP S → VP NP → Pronoun NP → Proper-Noun NP → Det Nominal Nominal → Noun Nominal → Nominal Noun Nominal → Nominal PP VP → Verb VP → Verb NP VP → VP PP PP → Prep NP	Det → the | a | that | this Noun → book | flight | meal | money Verb → book | include | prefer Pronoun → I | he | she | me Proper-Noun → Houston | NWA Aux → does Prep → from | to | on | near | through

b) Suppose the reputation of a university depends on it's academic excellence and infrastructure. These attributes are graded as poor, average, or good, with specific weights and degrees of belief assigned to each evaluation grade for assessment. 3 Ap 4

Evaluation Grade H_1=poor, H_2= average, H_3= good	Weight	Belief		
	Wi	$ß_{1,I}$	$ß_{2,I}$	$ß_{3,I}$
Accuracy	0.35	0.4	0.5	0
Reputation	0.65	0.1	0.75	0.15

Calculate the remaining belief and probability mass of the following attributes using DS theory.

Evaluation Grade H_1=poor, H_2= average, H_3= good	Belief	Probability mass					
	$ß_H$	$m_{1,I}$	$m_{2,I}$	$m_{3,I}$	$m_{H,I}$	$\bar{m}_{H,I}$	$\bar{\bar{m}}_{H,I}$
Accuracy							
Reputation							

c) What do you mean by uncertainty? What are the different sources of uncertainty. 3 U 2

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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 20__

Course Code: CSE 3635 Course Title: Artificial Intelligence

Total marks: 50

Time: 2 hours 30 minutes

The figures in the right-hand margin indicate full marks.

Course Outcomes and Bloom's Levels are mentioned in an additional Column

	Course Outcomes (COs) of the Questions
CO1	Understand the modern view of AI as the study of agents that receive precepts from the environment and perform actions.
CO2	Demonstrate awareness of informed search and exploration methods.
CO3	Apply AI techniques for knowledge representation, planning and uncertainty management.
CO4	Analyze different decision making and learning methods.

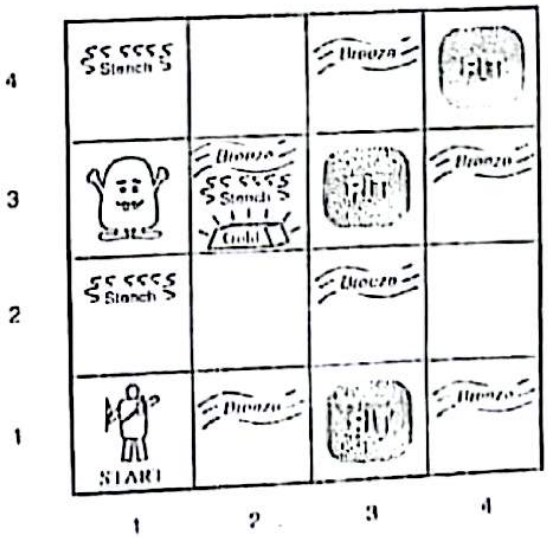
	Bloom's Levels of the Questions					
S	Remember	Understand	Apply	Analyze	Evaluate	Create
M	Remembr	Understand	Apply	Analyze	Evaluate	Create

Part A CLO BL

[Answer the following questions]

1. a) What is knowledge-base? Draw and discuss the architecture of a knowledge-based agent. CO3 U 1+2

   b) The figure below shows a Wumpus World game, where the agent starts from location [1,1]. Briefly discuss with diagrams how the agent generates new knowledge through perceiving the environment and choosing his next move. CO3 Ap 5



c) State the ontological and epistemological commitment of the propositional, first-order, probability theory, and fuzzy logic. CO2 U 2

OR,

1. a) What is knowledge-base? Write the tasks of *TELL* and *ASK function in a* knowledge based system. CO2 U 3

b) In a knowledge-based system, the following rules are given: CO Ap 5

Rule 1: A breeze in cell B22 implies that there is a pit in either cell P21, P23, P12, or P32.

Rule 2: There is no breeze in cell B22.

Using logical rules such as biconditional elimination, modus ponens, disjunction elimination, and other necessary logical rules, determine

$P_{21} \wedge \neg P_{23} \wedge \neg P_{12} \wedge \neg P_{32}$

c) How would you model the speed of a vehicle (65km/h) in both fuzzy and crisp sets? CO2 U 2

2. a) Differentiate between forward chaining and backward chaining. In what aspects forward chaining should be used? CO2 U 2+2

b) Write down the top-down and bottom up parse tree for the sentence "Book the flight through Houston" using the following CFG CO2 Ap 3

- Grammar
- S → NP VP
- S → Aux NP VP
- S → VP
- NP → Pronoun
- NP → Proper-Noun
- NP → Det Nominal
- Nominal → Noun
- Nominal → Nominal Noun
- Nominal → Nominal PP
- VP → Verb
- VP → Verb NP
- VP → VP PP
- PP → Prep NP

- Lexicon
- Det → that | this | a
- Noun → book | flight | meal | money
- Verb → book | include | prefer
- Pronoun → I | he | she | me
- Proper-Noun → Houston | NWA
- Aux → does
- Prep → from | to | on | near | through

c) Explain with example why universal quantification elimination is easy while existential quantification elimination is not that much easy CO3 U 3

Part B
[Answer the following questions]

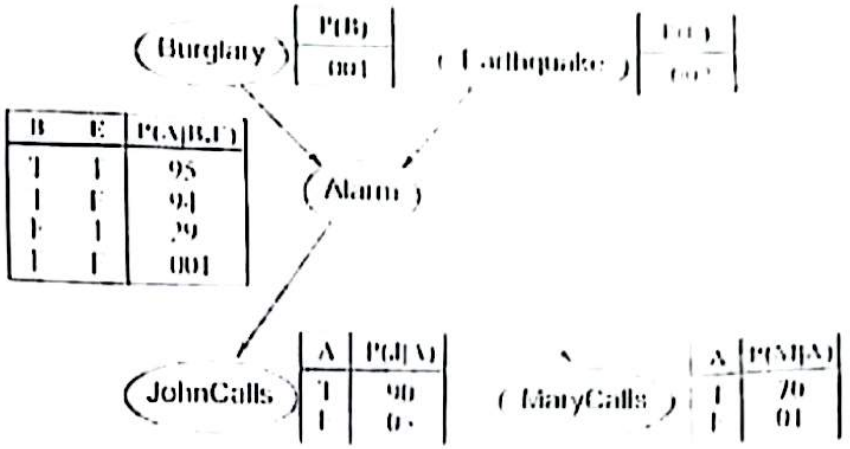
3. a) Explain the Bayes Theorem. CO4 U 2

3. b) A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 40% of the time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is 1/50000, and the prior probability that any patient has a stiff neck is 1/25. Find the probability of patients with a stiff neck having meningitis. CO4 Ap 3

3. c) A Bayesian network (Figure 1), showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J, and M stand for Burglary, Earthquake, Alarm John Calls, and MaryCalls, CO4 Ap 5

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respectively. The Independent conditional probability helps us to write in a simplified way the joint distribution P(B, E, A, J, M).



i) Express the joint distribution P(B, E, A, J, M) in terms of the conditional probabilities (and independencies) expressed in the Bayesian Network above.

ii) Probability of the event that the alarm has sounded but neither a burglary nor an earthquake has occurred and both Mary and John call.

iii) Probability of the event that the alarm has sounded and Burglary has occurred, an earthquake has not occurred, and both Mary and John call

4. a) The following probability distributions are given: CO3 Up 5

$P(W)$

R	P
sun	0.8
rain	0.2

$P(D|W)$

D	W	P
wet	sun	0.1
dry	sun	0.9
wet	rain	0.7
dry	rain	0.3

How Bayes theorem find P(W | dry) ?

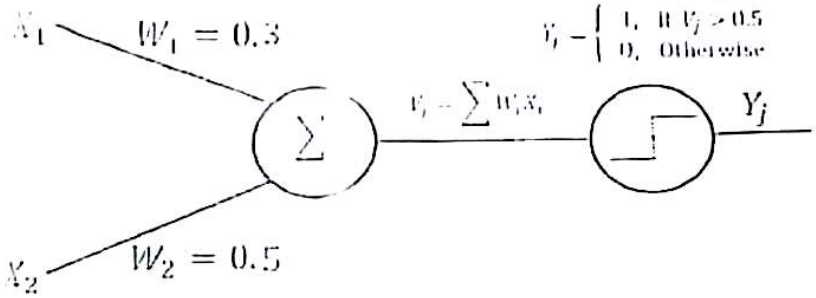
b) Suppose the reputation of a university depends on it's academic excellence and infrastructure. These attributes are graded as poor, average, or good, with specific weights and degrees of belief assigned to each evaluation grade for assessment. CO3 Up 5

Evaluation Grade H_1 poor; H_2= average, H_3= good	Weight	Belief		
	Wi	B_{1i}	B_{2i}	B_{3i}
Accuracy	0.35	0.4	0.5	0
Reputation	0.65	0.1	0.75	0.15

Calculate the remaining belief and probability mass of the following attributes using DS theory.

Evaluation Grade H_1=poor; H_2= average, H_3= good	Belief	Probability mass					
	B_H	m_{H1}	m_{H2}	$m_H i$	m_{H3}	m_H	m_H
Accuracy							
Reputation							

5. a) Consider the following neural network model with a learning rate of 0.2. CO 4



Demonstrate the learning process of the OR operation as shown below using the delta rule.

b) How do single-layer recurrent neural networks differ from multilayer feedforward neural networks? CO 3

c) What is natural language processing (NLP)? Briefly discuss the steps through which NLP is conducted. CO 3

OR,

5. a) What is natural language processing (NLP)? Describe the different phases of natural language processing (NLP) with example. CO 4

b) "*Fuzzy set is the generalized version of crisp set*"- Do you agree with the statement? Justify your answer. CO 3

c) What do you mean by uncertainty? What are the different sources of uncertainty. CO 3

International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 2022

Course Code: CSE 3635 Course Title: Artificial Intelligence

Total marks: 50

Time: 2 hours 30 minutes

[Answer the following questions. The figures in the right hand margin indicate full marks.]

Group A

1. a) The following knowledge base R1, R2, R3, R4, R5 has given below. CO2 4

R1: $\neg P_{1,1}$

R2: $B_{1,1} \leftrightarrow (P_{2,1} \vee P_{1,2})$

R3: $B_{2,1} \leftrightarrow (P_{1,1} \vee P_{2,2} \vee P_{3,1})$

R4: $\neg B_{1,1}$

R5: $B_{2,1}$

Show that $\neg P_{1,2}$ is true

Or,

A statement is given CO2 4

"As per the law, it is a crime for an American to sell weapons to hostile nations. Country A, an enemy of America, has some missiles, and all the missiles were sold to it by Robert, who is an American citizen."

Using Forward chaining method prove that "Robert is criminal."

b) Briefly explain why FOL is considered as the generalization of PL. CO1 2

Or,

How a Bayes' net encodes a joint distribution? CO1 2

c) What is Conjunctive Normal Form (CNF)? Convert the following of sentence to CNF form: CO3 4

$B_{1,1} \leftrightarrow (P_{2,1} \vee P_{1,2})$

Or,

The probability of some facts are given below CO3 4

S	T	W	P
summer	hot	sun	0.30
summer	hot	rain	0.05
summer	cold	sun	0.10
summer	cold	Rain	0.05
winter	hot	sun	0.10
winter	hot	rain	0.05
winter	cold	sun	0.15
winter	cold	rain	0.20

Calculate the following queries using inference by enumeration.

i. P(W)

ii. P(W | winter)

iii. P(W | winter, hot)

2. a) What is knowledge-base? Draw the architecture of a knowledge-based agent. CO1 3

b) Determine whether the following sentences is (i) Satisfiable (ii) Contradictory (iii) Valid CO2 3

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$(P \vee Q) \wedge (P \vee \sim Q)$

c) Represent the following sentences in First-Order-Logic: CO2 4

i. Every gardener likes the sun
ii. You can fool some of the people all of the time
iii. All purple mushrooms are poisonous
iv. Clinton is not tall.

Group B

3/

a) What is Baye's theorem? CO1 2

b) A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 40% of time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is 1/50000, and the prior probability that any patient has a stiff neck is 1/25. Find the probability of patients with a stiff neck to have meningitis. CO2 2

c) A Bayesian network (Figure 1), showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J and M stand for Burglary, Earthquake, Alarm John Calls, and MaryCalls, respectively. The Independent conditional probability help us to write in a simplified way the joint distribution P (B, E, A, J, M). CO3 6

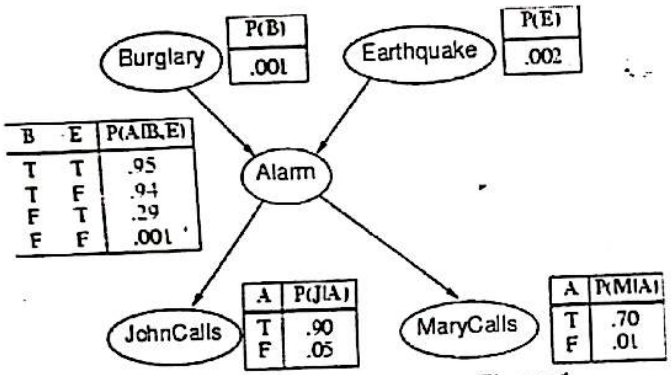


Figure 1

i) Express the joint distribution P (B, E, A, J, M) in terms of the conditional probabilities (and independencies) expressed in the Bayesian Network above.
ii) Probability of the event that the alarm has sounded but neither a Burglary nor an earthquake has occurred and both Mary and John call.
iii) Probability of the event that the alarm has sounded and Burglary has occurred, an earthquake has not occurred and both Mary and John call.

4. a) Compare artificial and biological networks. CO1 2

b) Consider a single neuron with two inputs X1 and X2, and output Y, learning the inclusive AND operation. The truth table is given below. CO3 8

Case	X1	X2	Z
1	0	0	0
2	0	1	0
3	1	0	0
4	1	1	1

Develop an ANN model using Delta rule. Where,
Threshold, T=0.5
Learning rate, alpha=0.2
Initial weights W1=0.1 and W2=0.5

Or,

a) What is Dempster-Shafer theory? CO1 2

b) Suppose the intrinsic quality of a product depends on it's accuracy and reputation. These attributes can be assessed according to the evaluation grades poor, average or good. For each attribute the weights and the degree of belief of each evaluation grades are given below: CO3 8

Evaluation Grade H_1=poor, H_2= average, H_3= good	Weight	Belief		
	Wi	$ß_{1,i}$	$ß_{2,i}$	$ß_{3,i}$
Accuracy	0.35	0.4	0.5	0
Reputation	0.65	0.1	0.75	0.15

Calculate the remaining belief and probability mass of the following attributes using DS theory.

Evaluation Grade H_1=poor, H_2= average, H_3= good	Belief	Probability mass					
	$ß_H$	$m_{1,i}$	$m_{2,i}$	$m_{3,i}$	$m_{H,i}$	$\bar{m}_{H,i}$	$\tilde{m}_{H,i}$
Accuracy							
Reputation							

5 a) What is NLP? Write about the steps through which NLP is conducted. CO2 3

b) Write down the parse tree for the sentence "Kawsar wrote the program" using the following simple grammar for a fragment of English in Figure 1. CO3 4

S → NP VP
NP → the NP1
NP → PRO
NP → PN
NP → NP1
NP1 → ADJS N
ADJS → ε|ADJS ADJS
VP → V
VP → V NP
N → file | program
PN → Kawsar
PRO → I
ADJ → short | Long | fast
V → printed | Created | wrote

Figure 1. A simple grammar for a fragment of English

c) What is learning? What are the supervised learning, unsupervised learning and reinforcement learning? CO2 3

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International Islamic University Chittagong

Department of Computer Science and Engineering

B Sc. in CSE

Final Examination, Spring-2022

Course Code: CSE-3635 Course Title: Artificial Intelligence

Total marks: 50 Time: 2 hours and 30 minutes

[Answer the following questions. The figures in the right-hand margin indicate full marks.]

GROUP A

1. a) Define forward chaining and backward chaining. CO1 2

Or

Differentiate between Modus Ponens and Generalized Modus Ponens. CO1 2

b) Following are the rules and initial facts in the rule base and database of facts of a rule based expert system. CO2 2

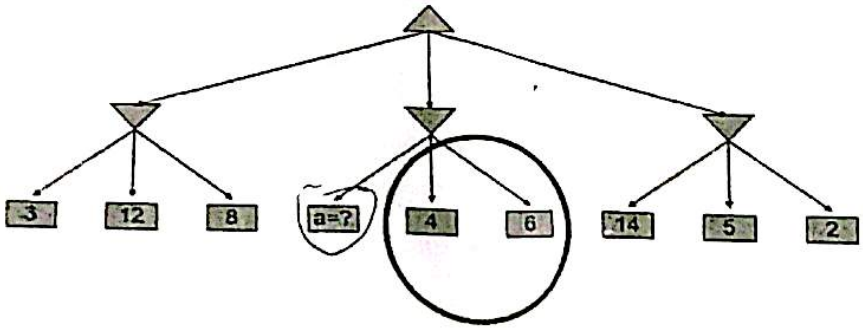
R1: IF A and C THEN F R2: IF A and E then G R3: IF B then E R4: IF G then D	Initial facts in the database A,B(both are true)

Prove if A and B true then D is true by using Forward chaining inference procedure.

c) Consider the following game tree. CO4 6

△ Max ▽ Min ▣ Terminal node

Notice that one terminal node marked 'a' has no value assigned (? marked). Suppose that the black circled node is pruned when you apply minimax search algorithm with alpha-beta pruning. Assign appropriate values to 'a' that will result in such a pruning.



2. a) Briefly discuss the steps through which NLP is conducted. CO1 2

b) Write down the top down and bottom-up parse tree for the sentence "Book the flight through Houston" using the following CFG. CO3 5

Grammar	Lexicon
S → NP VP	Det → the | a | that | this
S → Aux NP VP	Noun → book | flight | meal | money
S → VP 2	Verb → book | include | prefer
NP → Pronoun	Pronoun → I | he | she | me
NP → Proper-Noun	Proper-Noun → Houston | NWA
NP → Det Nominal	Aux → does
Nominal → Noun	Prep → from | to | on | near | through
Nominal → Nominal Noun	
Nominal → Nominal PP	
VP → Verb	
VP → Verb NP	
VP → VP PP	
PP → Prep NP	

Or Some Propositional Rules and a KB for the Wumpus world are given. CO3 5

Propositional rules:

(R1) $\neg S_{11} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$

(R2) $\neg S_{22} \rightarrow \neg W_{21} \wedge \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$

(R3) $\neg S_{12} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{22} \wedge \neg W_{13}$

(R4) $S_{12} \rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$

KB:

$\neg W_{11}$	$\neg S_{11}$	$\neg P_{11}$	$\neg B_{11}$	$\neg G_{11}$	V_{11}	OK_{11}
$\neg W_{12}$	—	$\neg P_{12}$	—	—	$\neg V_{12}$	OK_{12}
$\neg W_{21}$	$\neg S_{21}$	$\neg P_{21}$	B_{21}	$\neg G_{21}$	V_{21}	OK_{21}

Prove that Wumpus is in the room (1. 3)

c) Explain with example why universal quantification elimination is easy while existential quantification elimination is not that much easy. CO3 3

Or

Discuss a semantic model to address the following sentences, where the word "bank" carries different meaning. CO3 3

i. I went to the bank to observe sunset.
ii. I went to the bank to withdraw money.

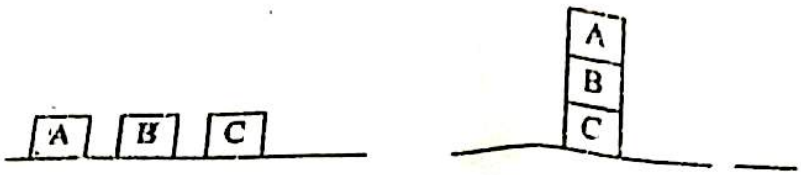
Group B

3. a) Write down the preconditions of the following operators:- CO3 4

i. UNSTACK (X, Y)
ii. STACK (X, Y)
iii. PICKUP(X)
iv. PUTDOWN (X)

b) Find out the desired sequence of actions for achieving the goal in the following example. CO3 6

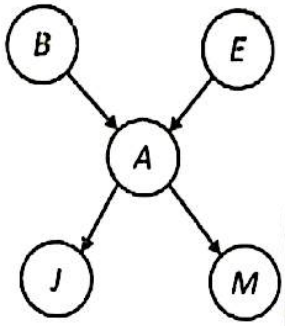
Total



Initial state Goal

4. a) What is Baye's theorem? CO1 2
Or What is inductive learning?

b) The following figure is a Bayesian network showing both the topology and conditional probability tables. CO4 8



B	P(B)
+b	0.001
-b	0.999

E	P(E)
+e	0.002
-e	0.998

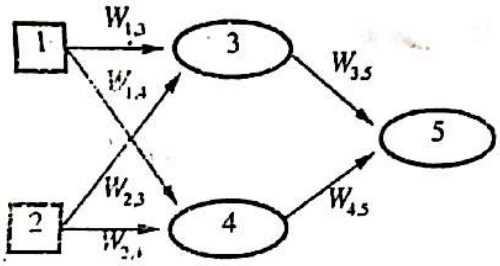
A	J	P(J|A)
+a	+j	0.9
+a	-j	0.1
-a	+j	0.05
-a	-j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	-m	0.3
-a	+m	0.01
-a	-m	0.99

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	-a	0.05
+b	-e	+a	0.94
+b	-e	-a	0.06
-b	+e	+a	0.29
-b	+e	-a	0.71
-b	-e	+a	0.001
-b	-e	-a	0.999

Calculate P(+b|+j,+m) using inference by enumeration.

OR Review the following neural network



I = 15. 2 = 8, $W_{1,3} = 0.5$, $W_{1,4} = 0.3$, $W_{2,3} = 0.2$, $W_{2,4} = 0.4$, $W_{3,5} = 0.5$, $W_{4,5} = 0.4$

I. Calculate hidden layer value

II. Compute the value of Z without the transfer function.

5. a) Differentiate between supervised and unsupervised learning with examples. CO2 2

b) The doctor knows that the disease meningitis causes the patients to have stiff neck, say, 40% of time. The doctor also knows the following facts, where, m: meningitis, S: stiff neck. CO3 6

$$P(+m) = 0.0001$$
$$P(+s|+m) = 0.8$$
$$P(+s|-m) = 0.01$$

Calculate P(+m|+s).

c) Let X is a set numbers near to zero.Define fuzzy and crisp sets for Z and show the triangular membership function curve in a graph. CO3 2

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International Islamic University Chittagong

Department of Computer Science and Engineering

B. Sc. in CSE Final Examination, Autumn 2021

Course Code: CSE 3635 Course Title: Artificial Intelligence

Total marks: 50

Time: 2 hours 30 minutes

Answer the following questions. The figures in the right hand margin indicate full marks.

Part A

1. a) Why First Order Predicate Logic is considered as the generalization of Propositional Logic? CO1 2

1. b) State the pros & the cons of Forward chaining and Backward chaining. CO1 4

Or

1. b) Consider the following axioms: CO1 4

P .....(1)
(P^Q) -->R .....(2)
(S v T) -->Q .....(3)
T .....(4)

(i) Convert the formulas into clause form
(i) Prove that whether R is true or not by using propositional resolution.

1 c) Consider the following facts: CO2 4

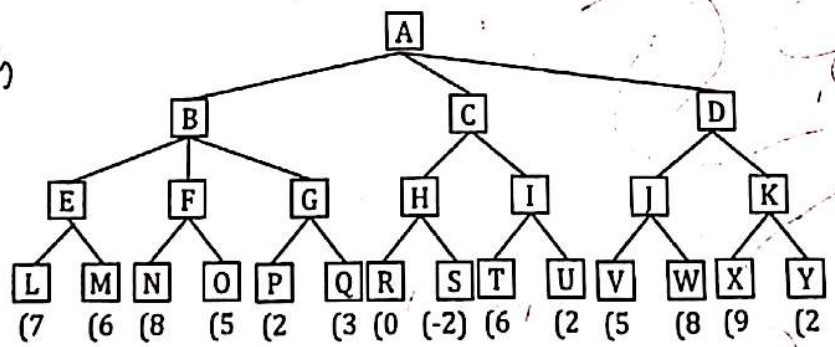
(1) Marcus was a man. (2) Marcus was a Pompeian. (3) Marcus was born in 40 A.D. (4) All men are mortal. (5) All Pompeian died when the volcano erupted in 79 A.D. (6) No mortal lives longer than 150 years. (7) It is now 2014. (8) Alive means not dead. (9) If someone dies, then he is dead at all later times.

-Translate these facts into well-formed formulas (wffs) in predicate logic.
-Answer the question "Is Marcus alive now?" using backward reasoning.
-Convert the formula into clause form.
-Prove that "Marcus is not alive now" using resolution.

2. a) Is the minimax procedure a depth-first or breadth-first search procedure? CO1 1

2. b) Describe the minimax search procedure. CO2 4

2. c)



CO4 5

(i) Suppose the first player is the maximizing player. What move should be chosen?

(ii) In the game tree, what nodes would not need to be examined using the alpha-beta pruning procedure?

Or

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2. c) Trace the constraint satisfaction procedure solving the following crypt-arithmetic problem: CO4 5

```
  S E N D
+ M O R E
-----------
M O N E Y
```

Initial State:
No two letters have the same value.
The sums of the digits must be as shown in the problem.

Part B

3. a) Identify planning in the AI. CO2 2
3. b) Demonstrate Goal Stack Planning. CO2 3

Or

3. b) Write down the preconditions of the following operators: (i) UNSTACK (A,B), (ii) STACK(A,B), (iii) PICKUP(A) and (iv) PUTDOWN(A). CO2 3
3. c) Develop an effective and complete plan using STRIPS approach (or any other approaches of your choice) to convert given initial state into goal state. CO2 5

4. a) What is Baye's theorem? CO1 1
4. b) Taking account of the example below, explain the concept of uncertainty. CO2 2
The doorbell rang at 12'0 clock in the midnight.
Was someone there at the door?
Did Karim wakeup?
4. c) A doctor knows that the disease meningitis causes the patient to have a stiff neck, say, 40% of time. The doctor also knows some unconditional facts: the prior probability that a patient has meningitis is 1/50000, and the prior probability that any patient has a stiff neck is 1/25. Find the probability of patients with a stiff neck to have meningitis. CO2 2.5
4. d) A Bayesian network (Figure 1), showing both the topology and the conditional probability tables (CPTs). In the CPTs, the letters B, E, A, J and M stand for Burglary, Earthquake, Alarm John Calls, and MaryCalls, respectively. The Independent conditional probability help us to write in a simplified way the joint distribution P (B, E, A, J, M). CO3 4.5

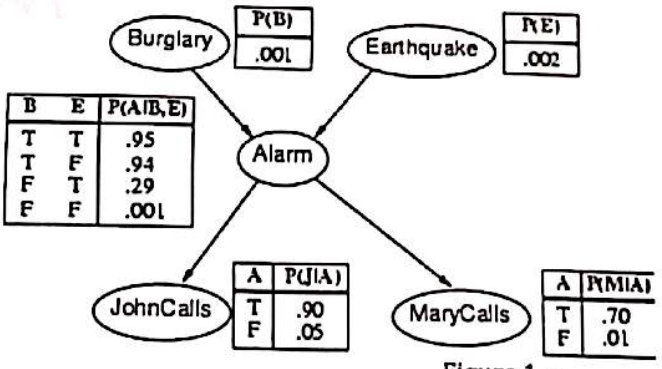


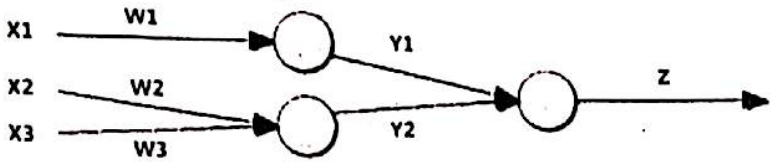
Figure 1

i) Express the joint distribution P (F, E, A, Y, S) in terms of the conditional probabilities (and independencies) expressed in the Bayesian Network above.
ii) Probability of the event that the alarm has sounded but neither a Burglary nor an earthquake has occurred and both Mary and John call.
iii) Probability of the event that the alarm has sounded and Burglary has occurred, an earthquake has not occurred and both Mary and John call.

5. a) Compare artificial and biological networks. What aspects of biological networks are not CO2

mimicked by artificial ones? What aspects are similar?

4.	b)	Write down the steps through which NLP is conducted.	CO1	3
5.	c)	Review this neural network and compute Z.	CO4	5



Where X1 = 15, X2 = 8, X3 = 14, W1 = 0.6, W2 = 0.3, W3 = 0.1, weight for Y1 = 0.6, weight for Y2 = 0.45

a) Compute the value of Z without a transfer function

b) Compare the value of Z with a threshold function. If the value is 5 or less, call it 0; otherwise call it 1.

Compute the value of Z with the sigmoid transfer function used at all neurons.

Or,

5	a)	Give your realization on inductive learning method with necessary example.	CO2	2
5	b)	Analyze the syntax and semantics of Bayesain Belief Network.	CO1	4
5	c)	Interpret how domain knowledge is represented in Expert System. Contrast two essential capabilities of an Expert System.	CO4	4

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